PRANAV JINDAL

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EDUCATION

Bachelor of Technology in Computer Science and Engineering Graduated **August 2025**
Vellore Institute of Technology **8.22 CGPA**

TECHNICAL SKILLS

Programming: C++, Go, Python, Java, SQL

Backend & Systems: Linux, REST APIs, Distributed Systems, Concurrency, Event-Driven Systems, Docker, PostgreSQL

Engineering & Testing: Git, CI/CD, Unit Testing, Integration Testing, Load Testing, Performance Profiling, Log Analysis

AI & Data: PySpark, Scikit-learn, YOLO, TensorFlow Lite, Depth Anything, GeoSAM

WORK EXPERIENCE

Indian Institute of Technology Bombay, Mumbai: Project Research Associate [On-Site] **October 2025 – Present**

- Built asynchronous C/C++ backend workflows with queue processing, state management, and service communication.
- Designed distributed workflows with request correlation, idempotency, concurrency safeguards, retries, and validation.
- Built a Go-based closed-loop load-testing framework using gNBSIM to simulate concurrent clients and benchmark workflows.
- Built performance instrumentation and automated testing, benchmarking 111,679 cycles in 300 seconds at 372.3 cycles/sec.

Indian Institute of Technology Roorkee, Roorkee: Project Research Associate [On-Site] **August 2025 – September 2025**

- Automated ADAS camera calibration using Depth Anything, reducing manual setup effort by 50%.
- Fine-tuned YOLO models on NVIDIA Jetson for real-time two-wheeler ADAS edge inference.
- Improved detection latency and reliability for ADAS deployment in resource-constrained embedded AI environments.
- Built Raspberry Pi telemetry pipelines for remote CPU, memory, and device-level monitoring.

Centre for Artificial Intelligence and Robotics, DRDO, Bangalore: Project Trainee [On-Site] **March 2025 – May 2025**

- Built a Python-based satellite image segmentation pipeline, reducing manual annotation effort by 60%.
- Implemented GeoSAM workflows to generate accurate masks and geospatial boundaries from remote sensing data.
- Designed CPU-compatible geospatial processing workflows, reducing dependency on GPUs and GIS-specific tools.
- Processed spatial outputs using Rasterio, Shapely, and Leafmap to support downstream geospatial analysis.

SVGS IT Solutions, Noida: Software Engineer Intern [Hybrid] **May 2024 – November 2024**

- Built Python-based machine learning workflows for data preprocessing, feature engineering, model training, and evaluation.
- Enhanced model performance by 15–20% through data cleaning, tuning, validation, and workflow refinement.
- Created dashboards to monitor business KPIs, model outputs, performance trends, and stakeholder insights.
- Automated recurring analytical reports using reusable Python workflows, reducing manual reporting effort.

PROJECTS

PySpark Financial Monitoring and Regulatory Alert Pipeline April 2026

Built a PySpark-based AML transaction monitoring pipeline to process transaction files, validate records, flag suspicious activity, and store alerts in PostgreSQL, simulating a BFSI-style workflow with regulatory-inspired controls and data quality checks.

- Built a PySpark batch ETL pipeline to extract, validate, transform, and load financial transaction data.
- Validated transaction records using Pydantic and separated invalid records into quarantined outputs.
- Implemented Spark DataFrame rules to flag high-value transfers and high-risk country transactions.
- Used Docker Compose with PostgreSQL to simulate a production-style financial data workflow.

PDF IntelliExtract - Automated PDF Data Extraction Platform January 2025

Built a full-stack PDF data extraction system using React, Node.js, Flask, and Python to parse documents and auto-fill structured form fields.

- Built a React-based interface for PDF uploads and automatic population of extracted user information.
- Developed a Node.js and Express API using Multer to process uploads and route documents to the extraction service.
- Implemented a Flask-based Python service using pdfplumber and regex to extract various structured fields from PDFs.
- Integrated frontend, backend, and document-processing services through REST APIs for end-to-end PDF processing.

LEADERSHIP & ACTIVITIES

Data Science Club, VIT Bhopal, : Co-Lead | June 2023

- Led a 25-member team to organize 5+ technical events, improving student engagement by 70%.
- Mentored 20+ students in Python, machine learning fundamentals, and project-based learning.